

Lecture 2 Point-referenced Data Models

Shiwei Lan¹

¹School of Mathematical and Statistical Sciences
Arizona State University

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Point-referenced

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Spatial
Problems

Elements of
point-referenced
modeling

Spatial process
model

Exploratory data
analysis (EDA)

Classical spatial
prediction

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- 2 Elements of point-referenced modeling
- 3 Spatial process model
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RISING AND FALLING NEW CORONAVIRUS CASES CHANGE IN DAILY NUMBER OF NEW CASES

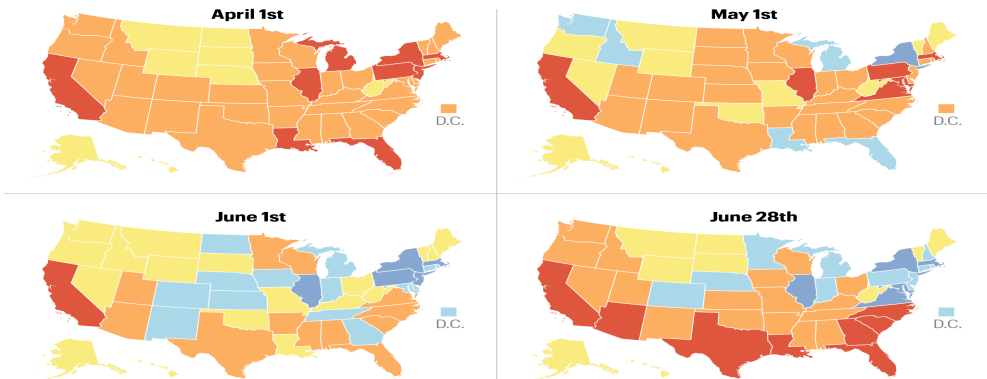
STRONG INCREASE

INCREASE

FLAT

DECREASE

STRONG DECREASE



SEVEN-DAY AVERAGE OF NEW CASES. "STRONG" CHANGE: IN EXCESS OF 500 CASES; "FLAT": +/- 25
 SOURCE: N.Y. TIMES COMPILATION OF STATE AND LOCAL GOVERNMENTS AND HEALTH DEPARTMENTS DATA

FORTUNE

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Assume the spatial process $Y(\mathbf{s})$ has a mean, $\mu(s) = E[Y(s)]$, and the variance of $Y(\mathbf{s})$ exists for all $\mathbf{s} \in D$.

Definition (Gaussian Process)

The process $Y(\mathbf{s})$ is said to be Gaussian if, for any $n \geq 1$ and any set of sites $\{\mathbf{s}_1, \dots, \mathbf{s}_n\}$, $\mathbf{Y} = \{Y(\mathbf{s}_1), \dots, Y(\mathbf{s}_n)\}$ has a multivariate normal distribution.

Definition ((strict) stationarity)

A process is said to be strictly stationary if, for any given $n \geq 1$, any set of n sites $\{\mathbf{s}_1, \dots, \mathbf{s}_n\}$ and any $\mathbf{h} \in \mathbb{R}^r$, the distribution of $(Y(\mathbf{s}_1), \dots, Y(\mathbf{s}_n))$ is the same as that of $(Y(\mathbf{s}_1 + \mathbf{h}), \dots, Y(\mathbf{s}_n + \mathbf{h}))$.

Definition (weak stationarity)

A spatial process $Y(\mathbf{s})$ is said to be weakly stationary if $\mu(\mathbf{s}) \equiv \mu$ (?) and

$$\text{Cov}(Y(\mathbf{s}), Y(\mathbf{s} + \mathbf{h})) = C(\mathbf{h}) \quad (1)$$

for all $\mathbf{h} \in \mathbb{R}^r$ such that $\mathbf{s}, \mathbf{s} + \mathbf{h} \in D$.

- Stationarity $\Rightarrow$ weak stationarity? Weak stationarity $\Rightarrow$ stationarity?
- How about Gaussian process?

Definition (intrinsic stationarity)

A spatial process $Y(\mathbf{s})$ is said to be *intrinsically stationary* if $E[Y(\mathbf{s} + \mathbf{h}) - Y(\mathbf{s})] = 0$ and

$$E[Y(\mathbf{s} + \mathbf{h}) - Y(\mathbf{s})]^2 = \text{Var}(Y(\mathbf{s} + \mathbf{h}) - Y(\mathbf{s})) = 2\gamma(\mathbf{h}) \quad (2)$$

- $2\gamma(\mathbf{h})$ is called *variogram* and $\gamma(\mathbf{h})$ is named *semivariogram*.
- We have

$$\gamma(\mathbf{h}) = C(0) - C(\mathbf{h}) \quad (3)$$

- Can we recover C from γ ?
- Weak stationarity $\Rightarrow$ intrinsic stationarity? Intrinsic stationarity $\Rightarrow$ weak stationarity?

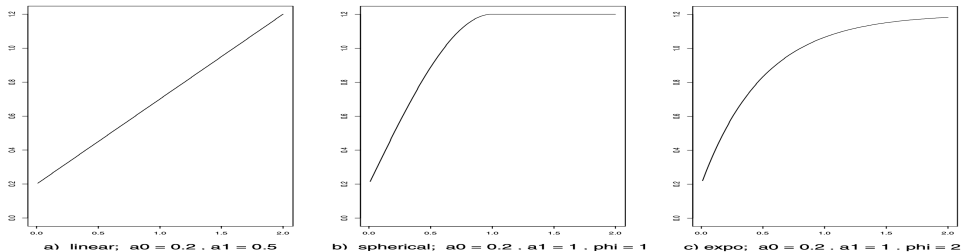


Figure 2.1 *Theoretical semivariograms for three models: (a) linear, (b) spherical, and (c) exponential.*

For any set of locations $\{\mathbf{s}_1, \dots, \mathbf{s}_n\}$ and any constants $a_1, \dots, a_n$ such that $\sum_i a_i = 0$.

$$\sum_i \sum_j a_i a_j \gamma(\mathbf{s}_i - \mathbf{s}_j) \leq 0 \quad (4)$$

Definition (isotropy)

A spatial process $Y(\mathbf{s})$ is said to be isotropic if the semivariogram function $\gamma(\mathbf{h})$ depends upon the separation vector only through its length $\|\mathbf{h}\|$, i.e.

$$\gamma(\mathbf{h}) = \gamma(\|\mathbf{h}\|) \quad (5)$$

Otherwise we say it is anisotropic.

- Linear:

$$\gamma(t) = \begin{cases} \tau^2 + \sigma^2 t & \text{if } t > 0, \tau^2 > 0, \sigma^2 > 0 \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

- Spherical:

$$\gamma(t) = \begin{cases} \tau^2 + \sigma^2 & \text{if } t \geq 1/\phi \\ \tau^2 + \sigma^2 \left\{ \frac{3\phi t}{2} - \frac{1}{2}(\phi t)^3 \right\} & \text{if } 0 < t \leq 1/\phi \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

- Exponential:

$$\gamma(t) = \begin{cases} \tau^2 + \sigma^2(1 - \exp(-\phi t)) & \text{if } t > 0 \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

- Gaussian:

$$\gamma(t) = \begin{cases} \tau^2 + \sigma^2(1 - \exp(-\phi^2 t^2)) & \text{if } t > 0 \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

model	Variogram, $\gamma(t)$
Linear	$\gamma(t) = \begin{cases} \tau^2 + \sigma^2 t & \text{if } t > 0 \\ 0 & \text{otherwise} \end{cases}$
Spherical	$\gamma(t) = \begin{cases} \tau^2 + \sigma^2 \left[\frac{3}{2} \phi t - \frac{1}{2} (\phi t)^3 \right] & \text{if } t \geq 1/\phi \\ 0 & \text{if } 0 < t \leq 1/\phi \\ 0 & \text{otherwise} \end{cases}$
Exponential	$\gamma(t) = \begin{cases} \tau^2 + \sigma^2 (1 - \exp(-\phi t)) & \text{if } t > 0 \\ 0 & \text{otherwise} \end{cases}$
Powered exponential	$\gamma(t) = \begin{cases} \tau^2 + \sigma^2 (1 - \exp(- \phi t ^p)) & \text{if } t > 0 \\ 0 & \text{otherwise} \end{cases}$
Gaussian	$\gamma(t) = \begin{cases} \tau^2 + \sigma^2 (1 - \exp(-\phi^2 t^2)) & \text{if } t > 0 \\ 0 & \text{otherwise} \end{cases}$
Rational quadratic	$\gamma(t) = \begin{cases} \tau^2 + \frac{\sigma^2 t^2}{(1 + \phi t^2)} & \text{if } t > 0 \\ 0 & \text{otherwise} \end{cases}$
Wave	$\gamma(t) = \begin{cases} \tau^2 + \sigma^2 (1 - \frac{\sin(\phi t)}{\phi t}) & \text{if } t > 0 \\ 0 & \text{otherwise} \end{cases}$
Power law	$\gamma(t) = \begin{cases} \tau^2 + \sigma^2 t^\lambda & \text{if } t > 0 \\ 0 & \text{otherwise} \end{cases}$
Matérn	$\gamma(t) = \begin{cases} \tau^2 + \sigma^2 \left[1 - \frac{(2\sqrt{\nu} t \phi)^\nu}{2^{\nu-1} \Gamma(\nu)} K_\nu(2\sqrt{\nu} t \phi) \right] & \text{if } t > 0 \\ 0 & \text{otherwise} \end{cases}$
Matérn at $\nu = 3/2$	$\gamma(t) = \begin{cases} \tau^2 + \sigma^2 [1 - (1 + \phi t) \exp(-\phi t)] & \text{if } t > 0 \\ 0 & \text{otherwise} \end{cases}$

Table 2.2 Summary of variograms for common parametric isotropic models.12/29

Model	Covariance function, $C(t)$
Linear	$C(t)$ does not exist
Spherical	$C(t) = \begin{cases} 0 & \text{if } t \geq 1/\phi \\ \frac{\sigma^2}{\tau^2 + \sigma^2} \left[1 - \frac{3}{2}\phi t + \frac{1}{2}(\phi t)^3 \right] & \text{if } 0 < t \leq 1/\phi \\ \text{otherwise} & \text{otherwise} \end{cases}$
Exponential	$C(t) = \begin{cases} \frac{\sigma^2}{\tau^2 + \sigma^2} \exp(-\phi t) & \text{if } t > 0 \\ \text{otherwise} & \text{otherwise} \end{cases}$
Powered exponential	$C(t) = \begin{cases} \frac{\sigma^2}{\tau^2 + \sigma^2} \exp(- \phi t ^p) & \text{if } t > 0 \\ \text{otherwise} & \text{otherwise} \end{cases}$
Gaussian	$C(t) = \begin{cases} \frac{\sigma^2}{\tau^2 + \sigma^2} \exp(-\phi^2 t^2) & \text{if } t > 0 \\ \text{otherwise} & \text{otherwise} \end{cases}$
Rational quadratic	$C(t) = \begin{cases} \frac{\sigma^2}{\tau^2 + \sigma^2} \left(1 - \frac{t^2}{(1 + \phi t^2)} \right) & \text{if } t > 0 \\ \text{otherwise} & \text{otherwise} \end{cases}$
Wave	$C(t) = \begin{cases} \frac{\sigma^2 \sin(\phi t)}{\phi t} & \text{if } t > 0 \\ \frac{\sigma^2}{\tau^2 + \sigma^2} & \text{otherwise} \end{cases}$
Power law	$C(t)$ does not exist
Matérn	$C(t) = \begin{cases} \frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)} (2\sqrt{\nu}t\phi)^{\nu} K_{\nu}(2\sqrt{\nu}t\phi) & \text{if } t > 0 \\ \frac{\sigma^2}{\tau^2 + \sigma^2} & \text{otherwise} \end{cases}$
Matérn at $\nu = 3/2$	$C(t) = \begin{cases} \frac{\sigma^2}{\tau^2 + \sigma^2} (1 + \phi t) \exp(-\phi t) & \text{if } t > 0 \\ \text{otherwise} & \text{otherwise} \end{cases}$

Table 2.1 Summary of covariance functions (covariograms) for common parametric isotropic models.

- Matheron (1963) proposed the *empirical semivariogram* as an estimator

$$\hat{\gamma}(t) = \frac{1}{2N(t)} \sum_{(\mathbf{s}_i, \mathbf{s}_j) \in N(t)} [Y(\mathbf{s}_i) - Y(\mathbf{s}_j)]^2 \quad (10)$$

where for some grid $0 < t_1 < \cdots t_k$, let $I_k = (t_{k-1}, t_k)$,

$$N(t_k) = \{(\mathbf{s}_i, \mathbf{s}_j) : \|\mathbf{s}_i - \mathbf{s}_j\| \in I_k\}, \quad k = 1, \dots, K \quad (11)$$

- Cressie and Hawkins (1980) proposed a robustified estimate

$$\hat{\gamma}(t) = \frac{1}{2N(t)} \sum_{(\mathbf{s}_i, \mathbf{s}_j) \in N(t)} |Y(\mathbf{s}_i) - Y(\mathbf{s}_j)|^{1/2} \quad (12)$$

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- We require a 'valid' covariance $c(\mathbf{h}) = \text{Cov}(Y(\mathbf{s}), Y(\mathbf{s} + \mathbf{h}))$ such that for any finite sites $\mathbf{s}_1, \dots, \mathbf{s}_n$ and for any $a_1, \dots, a_n$,

$$\text{Var} \left[\sum_i a_i Y(\mathbf{s}_i) \right] = \sum_{i,j} a_{i,j} \text{Cov}(Y(\mathbf{s}_i), Y(\mathbf{s}_j)) = \sum_{i,j} a_{i,j} c(\mathbf{s}_i - \mathbf{s}_j) \geq 0 \quad (13)$$

- To verify the positivity of the covariance function, we have the following *Bochner's Theorem*.

Theorem

$c(\mathbf{h})$ is positive definite if and only if

$$c(\mathbf{h}) = \int \cos(\mathbf{w}^T \mathbf{h}) G(d\mathbf{w}) \quad (14)$$

where G is a bounded, positive, symmetric about 0 measure in $\mathbb{R}^r$.

- Note $c(\mathbf{0}) = \int G(d\mathbf{w})$ becomes a normalizing constant.
- $G(d\mathbf{w})/c(\mathbf{0})$ is referred to as the *spectral distribution* that induces $c(\mathbf{h})$.
- If $G(d\mathbf{w})$ has a density with respect to Lebesgue measure, i.e. $G(d\mathbf{w}) = g(\mathbf{w})d\mathbf{w}$, then $g(\mathbf{w})/c(\mathbf{0})$ is referred to as *spectral density*.
- Due to the symmetry of G around 0, we have $c(\mathbf{h})$ as a characteristic function of G :

$$c(\mathbf{h}) = \int \exp\{i\mathbf{w}^T \mathbf{h}\} G(d\mathbf{w}) \quad (15)$$

- Denote the Fourier transform of $c(\mathbf{h})$ as $\hat{c}(\mathbf{w}) = \int \exp\{-i\mathbf{w}^T \mathbf{h}\} c(\mathbf{h}) d\mathbf{h}$. Then we have the spectral density as $g(\mathbf{w}) = (2\pi)^{-r} \hat{c}(\mathbf{w})/c(\mathbf{0})$.

There are many ways to construct correlation functions from existing ones.

- *mixing*: Suppose $c_1, \dots, c_m$ are valid correlation function in $\mathbb{R}^r$, then $c(\mathbf{h}) = \sum_{i=1}^m p_i c_i(\mathbf{h})$ is also a valid correlation function for $p_i > 0$ and $\sum_{i=1}^m p_i = 1$.
- *product*: $c(\mathbf{h}) = \prod_{i=1}^m c_i(\mathbf{h})$ is also a valid correlation function.
- *convolution*: $c_{12}(\mathbf{h}) = \int c_1(\mathbf{h} - \mathbf{t}) c_2(\mathbf{t}) d\mathbf{t}$.

We could also construction anisotropic correlation c from isotropic one ρ :

$$c(\mathbf{s} - \mathbf{s}') = \sigma^2 \rho((\mathbf{s} - \mathbf{s}')^T B (\mathbf{s} - \mathbf{s}')) \quad (16)$$

where B is a positive definite matrix in $\mathbb{R}^r$. This is also called *geometric anisotropy*.

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- Exploratory data analysis (EDA) tools are routines for analyzing one- and two-sample data sets, regression studies, etc..
- For continuous data, the starting point is the *first law of geostatistics* that decomposes data into mean and error.
- EDA tools exam both first- and second-order behavior.

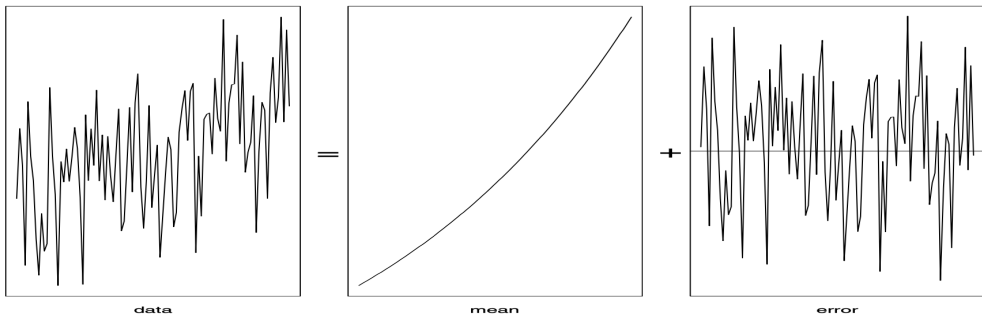


Figure 2.2 *Illustration of the first law of geostatistics.*

- To examine how *regular* the arrangement of the points is, we could use drop line, surface plot, or a smoothed summary contour boxplot.

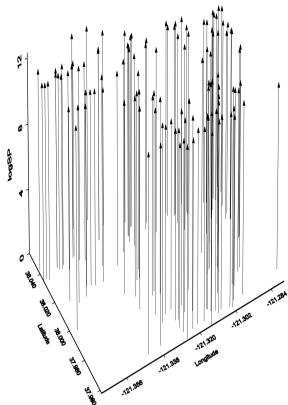


Figure 2.3 Illustrative three-dimensional "drop line" scatterplot, scallop data.

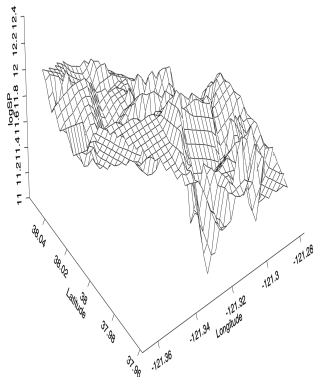
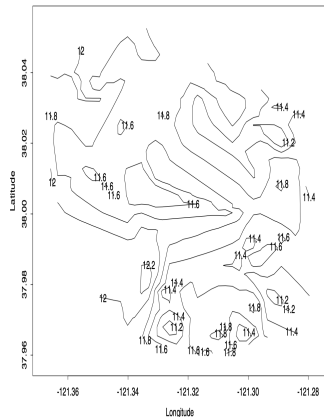


Figure 2.4 Illustrative three-dimensional surface ("perspective") plot, Stockton real estate data.



- To reveal more information about the trend, $\mu(s)$, we could use row boxplot, and column plot.

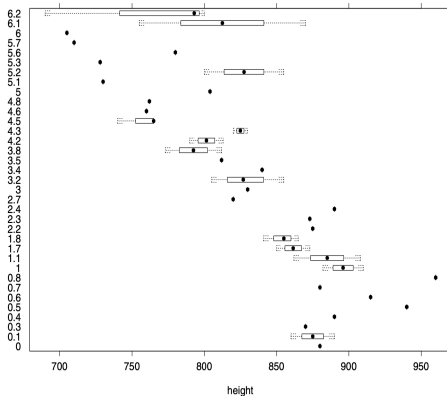


Figure 2.6 Illustrative row box plots, Diggle and Ribeiro (2002) surface elevation data.

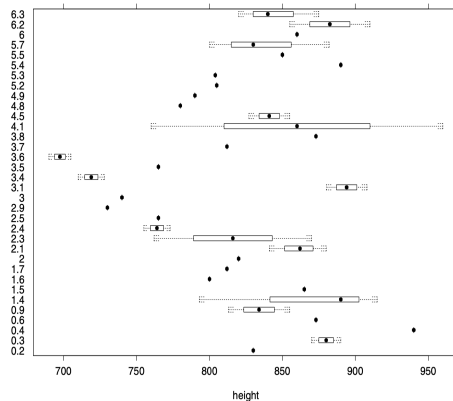


Figure 2.7 Illustrative column box plots, Diggle and Ribeiro (2002) surface elevation data.

- To access small-scale behavior, we could use an empirical (nonparametric) covariance estimate

$$\hat{c}(t_k) = \frac{1}{N_k} \sum_{(\mathbf{s}_i, \mathbf{s}_j) \in N(t_k)} (Y(\mathbf{s}_i) - \bar{Y})(Y(\mathbf{s}_j) - \bar{Y}) \quad (17)$$

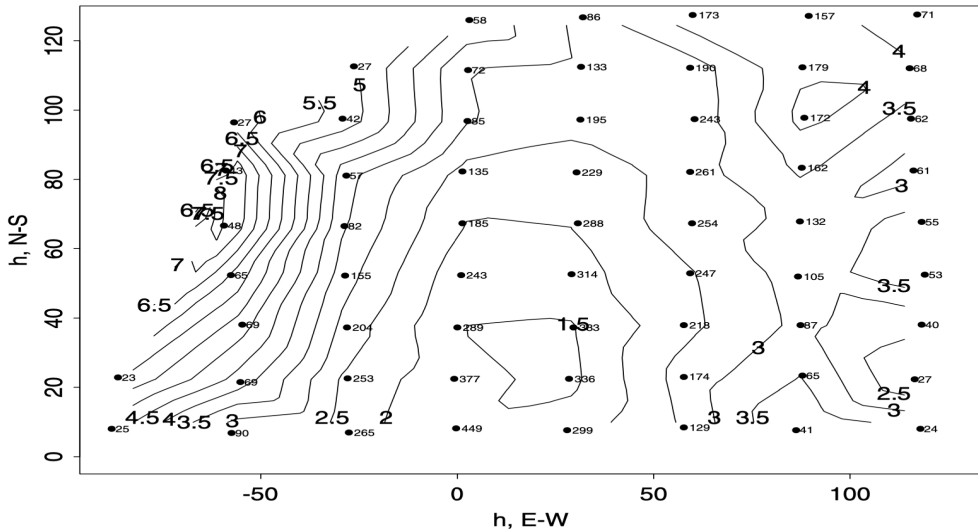
where $N(t_k) = \{(\mathbf{s}_i, \mathbf{s}_j) : \|\mathbf{s}_i - \mathbf{s}_j\| \in I_k\}$ for $k = 1, \dots, K$.

- On a regular grid or binning, we could create 'same-lag' scatter plots, i.e. $Y(\mathbf{s}_i + h\mathbf{e})$ vs $Y(\mathbf{s}_i)$ for a fixed h and a fixed $\mathbf{e}$ to reveal the presence of anisotropy and nonstationarity.
- We could use a 'slide-window' type of investigation for the trend and pattern. Suppose we attach a neighborhood to each point. We could compute the sample mean, sample variance or sample correlation coefficient using all the points in the neighborhood.

- To assess the anisotropy, we use *empirical semivariogram contour (ESC)* plot.
- For each of $\frac{N(N-1)}{2}$ pairs of sites in $\mathbb{R}^2$, calculate h_x and h_y , the separation distances along each axis. Restrict $h_y \geq 0$ and aggregate these distances into rectangular bins B_{ij} .
- Calculate empirical semivariogram values for (i,j) th bin:

$$\gamma_{ij}^* = \frac{1}{2N_{B_{ij}}} \sum_{(k,l): (\mathbf{s}_k - \mathbf{s}_l) \in B_{ij}} (Y(\mathbf{s}_k) - Y(\mathbf{s}_l))^2 \quad (18)$$

where $N_{B_{ij}}$ is the number of sites in bin B_{ij} .



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- *Kriging*, named by Matheron (1963), in honor of D.G. Krige (1951).
- Given observations of a random field $\mathbf{Y} = (Y(\mathbf{s}_1), \dots, Y(\mathbf{s}_n))$, how to predict the variable Y at a site $\mathbf{s}_0$ where it has not been observed?

- Suppose we model

$$\mathbf{Y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim N(\mathbf{0}, \boldsymbol{\Sigma}) \quad (19)$$

- We have

$$E[(Y(\mathbf{s}_0) - f(\mathbf{y}))^2 | \mathbf{y}] = E[(Y(\mathbf{s}_0) - E[Y(\mathbf{s}_0) | \mathbf{y}])^2 | \mathbf{y}] + (E[Y(\mathbf{s}_0) | \mathbf{y}] - f(\mathbf{y}))^2 \quad (20)$$

- $f(\mathbf{y}) = E[Y(\mathbf{s}_0) | \mathbf{y}]$, the posterior mean, is the best solution!

- Let

$$\Sigma = \sigma^2 H(\phi) + \tau^2 I \quad (21)$$

where $(H(\phi))_{i,j} = \rho(\phi; d_{ij})$.

- Using the property of multivariate normal $[Y(\mathbf{s}_0), \mathbf{y}]$, we obtain

$$\begin{aligned} E[Y(\mathbf{s}_0)|\mathbf{y}] &= \mathbf{x}_0^T \boldsymbol{\beta} + \boldsymbol{\gamma}^T \boldsymbol{\Sigma}^{-1}(\mathbf{y} - X\boldsymbol{\beta}) \\ \text{Var}[Y(\mathbf{s}_0)|\mathbf{y}] &= \sigma_0^2 + \tau^2 - \boldsymbol{\gamma}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\gamma} \end{aligned} \quad (22)$$

where $\mathbf{x}_0 = \mathbf{x}(\mathbf{s}_0)$, and $\boldsymbol{\gamma}^T = (\sigma^2 \rho(\phi; d_{01}), \dots, \sigma^2 \rho(\phi; d_{0n}))$.

- In practice, we need estimate parameters using data. Note we modify $f(\mathbf{y})$

$$\widehat{f(\mathbf{y})} = \mathbf{x}_0^T \widehat{\beta} + \widehat{\gamma}^T \widehat{\Sigma}^{-1} (\mathbf{y} - X \widehat{\beta}) \quad (23)$$

where $\widehat{\gamma} = (\hat{\sigma}^2 \rho(\hat{\phi}; d_{01}), \dots, \hat{\sigma}^2 \rho(\hat{\phi}; d_{0n}))$, $\widehat{\beta} = (X^T \widehat{\Sigma}^{-1} X)^{-1} X^T \widehat{\Sigma}^{-1} \mathbf{y}$, and $\widehat{\Sigma} = \hat{\sigma}^2 H(\hat{\phi})$.

- Thus $\widehat{f(\mathbf{y})}$ can be written as linear function of $\mathbf{y}$, $\widehat{f(\mathbf{y})} = \boldsymbol{\lambda}^T \mathbf{y}$, where

$$\boldsymbol{\lambda} = \widehat{\Sigma}^{-1} \widehat{\gamma} + \widehat{\Sigma}^{-1} X (X^T \widehat{\Sigma}^{-1} X)^{-1} (\mathbf{x}_0 - X^T \widehat{\Sigma}^{-1} \widehat{\gamma}) \quad (24)$$